

LAB 401 - Diabetes Mellitus and Glucose Testing

Source-faithful one-page wall chart from lecture No (2)-s.pdf

1. Core Definition

**Diabetes mellitus**

A state of diminished insulin action due to either decreased insulin production or decreased effectiveness of produced insulin.

**Type 1 DM**

Insulin dependent DM (IDDM), juvenile type, Type I DM.

**Type 2 DM**

Non insulin dependent DM (NIDDM), maturity onset DM, Type II DM.

**Other categories**

Secondary hyperglycemia: pancreatic diseases, Cushing, thyrotoxicosis, pheochromocytoma, acromegaly, stress, cerebrovascular accidents, drugs/hormones such as thiazide diuretics, oral contraceptives, corticosteroids, Down syndrome, Turner syndrome, and insulin receptor abnormalities.

2. Reference Ranges + Prediabetes

Test / state	Lecture value
Fasting plasma glucose	Normal: < 100 mg/dL after no caloric intake for at least 8 hours.
2-hour postprandial / OGTT	Normal: < 140 mg/dL after 75 g anhydrous glucose in non-pregnant adult; child dose 1.75 g/kg.
Random plasma glucose	Normal: < 200 mg/dL.
Impaired fasting glucose	Fasting 100-125 mg/dL with normal 2-hour post-load < 140 mg/dL.
Impaired glucose tolerance	Fasting below diabetic threshold, but OGTT 2-hour > 140 and < 200 mg/dL.
Gestational DM	First detection of DM is in pregnancy.

3. Diagnosis + Monitoring + Complications

**Random glucose**

>= 200 mg/dL plus classic symptoms of DM.

**Fasting glucose**

>= 126 mg/dL.

**2-hour OGTT**

>= 200 mg/dL.

**HbA1c**

>= 6.5%; added in 2010.

**Confirmation rule**

Any listed criterion is diagnostic, but in absence of hyperglycemia, confirm by repeating the test on a subsequent day.

**HbA1c**

Represents 80% of Hb A1, about 4-6% of Hb A. Assesses long-term control for 6-8 weeks; monitored routinely every 6 months.

**Fructosamine**

Short-term control monitor by glycated proteins, mainly albumin, for the previous 2-3 weeks.

**Metabolic complications**

Diabetic ketoacidosis; hyperosmolar non-ketotic hyperglycemic coma; lactic acidosis with or without hyperglycemia; hypoglycemic coma; chronic renal failure due to diabetic nephropathy.

4. Sample Precautions

**Specimen**

Serum or plasma free of hemolysis.

**Anticoagulants**

Heparin, EDTA, citrate, and oxalate cause no interference.

**Separation**

Serum must be promptly separated from whole coagulated blood to avoid glycolysis.

**Glycolysis effect**

Blood glucose decreases by 5-10 mg/dL in one hour if not separated.

**Stability**

Glucose is stable for 8 hours in separated non-hemolyzed serum at 25 C and 72 hours at 4 C.

5. Oral Glucose Tolerance Test (OGTT)

Part	Lecture details
Definition	Serial measurements of plasma glucose before and after a specific amount of glucose given orally.
Indications	Gestational DM; impaired glucose tolerance or impaired fasting glucose; unexplained nephropathy, retinopathy, neuropathy, renal glycosuria, or alimentary glycosuria.
ADA note	American Diabetes Association does not recommend routine use of OGTT.
Before test	Not in acutely ill, hospitalized, or bed-ridden patients. Stop drugs affecting glucose tolerance if possible. Three days unrestricted diet with at least 150 g carbohydrate/day and usual activity. Avoid heavy exercise.
Glucose load	Non-pregnant adult: 75 g. Pregnant female: 100 g. Child: 1.75 g/kg, max 75 g. Dissolve anhydrous glucose in 300 mL ice-chilled water and ingest over 5 minutes.
During test	Complete physical and mental rest. No eating or smoking. Patient seated or lying on right side to promote stomach emptying.
Blood sampling	Fasting sample, then 30, 60, 90, and 120 minutes after glucose. Pregnant female: fasting, 1 h, 2 h, 3 h.
Urine sampling	Fasting, 1-hour, and 2-hour post-load urine samples.

6. OGTT Interpretation Patterns

Pattern	Fasting	2-hour / curve	Other lecture clue
Normal OGTT	< 100 mg/dL	Peak <= 170 mg/dL; 2-hour < 140 mg/dL, usually within +/- 10% of fasting	All urine samples negative for glucose.
Diabetes mellitus	> 126 mg/dL	> 200 mg/dL	Diagnostic abnormal OGTT pattern.
Impaired glucose tolerance	< 126 mg/dL	> 140 and < 200 mg/dL	Prediabetes category.
Impaired fasting glucose	100-125 mg/dL	< 140 mg/dL	Prediabetes category.
Renal glucosuria	Variable	Curve may not match urine glucose	Pregnancy, Fanconi syndrome, heavy metal poisoning, renal tubular acidosis.
Alimentary glucosuria / lag curve	Variable	Post-gastrectomy style lag curve	Post-gastrectomy, advanced liver cirrhosis, thyrotoxicosis, apparently healthy person.
Flat curve	Variable	Flat response	Malabsorption, insulinoma, decreased insulin antagonist.
Gestational DM	95 mg/dL	1 h 180, 2 h 155, 3 h 140	Used for screening or diagnosis.

7. Lecture Admin + References

Clinical pathology / laboratory medicine. By Prof. Dr. Ahmed El Sayed Abd ALLAH El Tawil, Head of Clinical Pathology Department, Faculty of Medicine, MUST University. References listed in source: Adobe Stock diabetes complications search; iStock gestational diabetes photos; Diabetes Care article at care.diabetesjournals.org/content/26/4/1026.